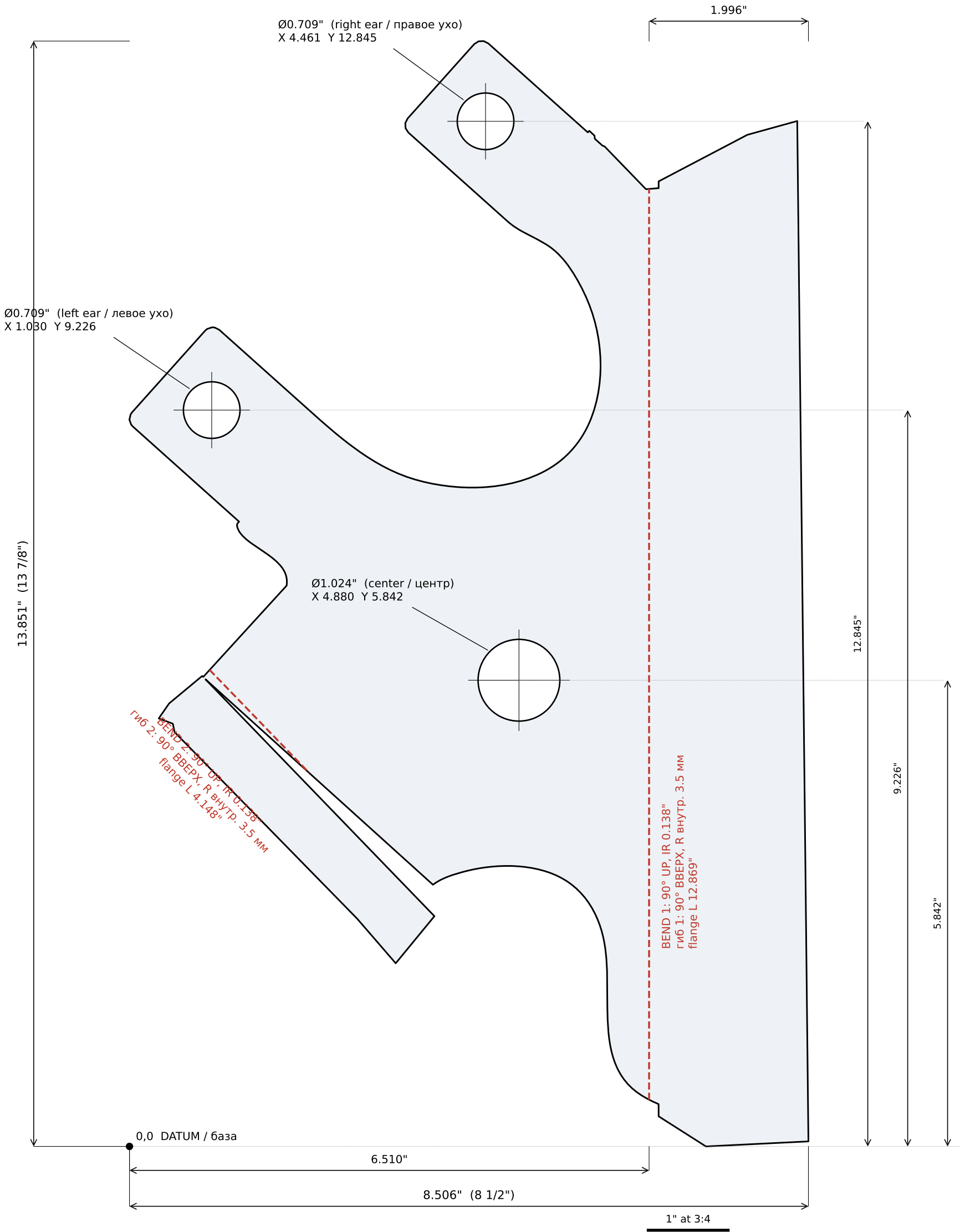


FR-1 — FLAT PATTERN / РАЗВЕРТКА (scale 3:4 / масштаб 3:4)

blank 8.506" × 13.851" · 3.5 mm (0.138") plate · 2 bends 90° up · IR 0.138"



NOTES / ПРИМЕЧАНИЯ

1. Cut from the DXF file FR-1 flat 3-5mm.dxf (1:1, inches). This sheet is for checking. Резать по DXF (1:1, дюймы); чертёж — для проверки.

2. Both bends 90° UP to the SAME side (toward the viewer), inside radius 0.138" (R = t = 3.5 mm). Обагиба 90° в одну сторону (к себе), R внутр. = 3.5 мм.

3. Flat pattern computed with K = 0.44, bend allowance 0.312", bend deduction 0.240" per bend.
If your tooling/K differs, keep the FINISHED sizes on sheet 2 and adjust the blank. Если ваш инструмент другой — держите готовые размеры (лист 2).

4. Red dashed lines = bend CENTER lines. Красный пунктир = средняя линиягиба.

5. Holes are laser cut in the flat (no drilling). Отверстия режутся лазером сразу.

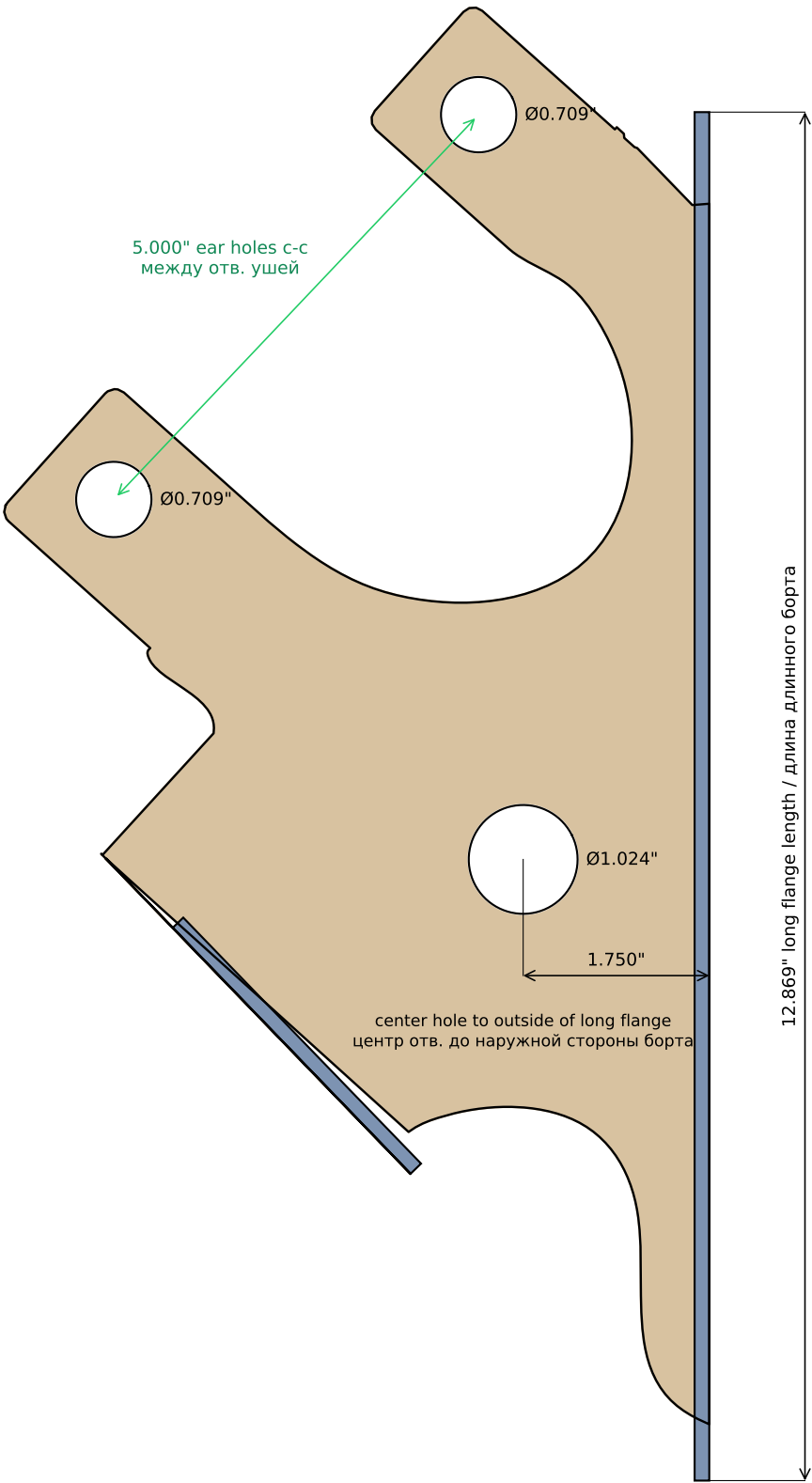
6. QTY 2: FR-1 as shown + FL-1 = SAME blank flipped over (bends to the other side = mirror part).
2 шт: FR-1 как на чертеже + FL-1 = та же заготовка, перевернуть (гнуть в другую сторону = зеркальная деталь).

7. Deburr all edges. Снять заусенцы. Curved edges: smooth arcs/splines from the DXF.

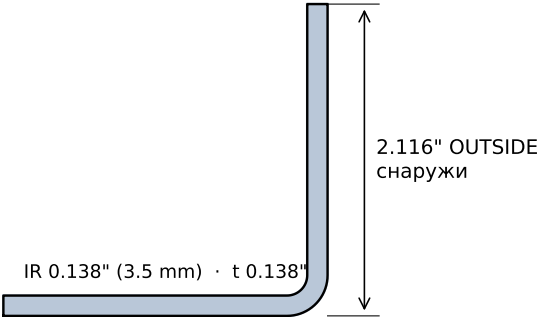
FR-1 · side plate / боковая пластина	A36 HR steel 3.5 mm (0.138")	inch / дюймы	Sienna WAV · 2026-10-02
FLAT PATTERN 3:4 / РАЗВЕРТКА 3:4	QTY 2 (FR-1 + FL-1 mirror)	cut ±0.010" · bend ±1°	SHEET 1 / 2

FR-1 — FINISHED PART / ГОТОВАЯ ДЕТАЛЬ

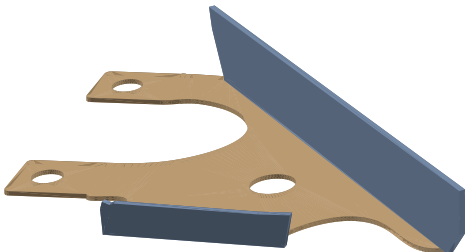
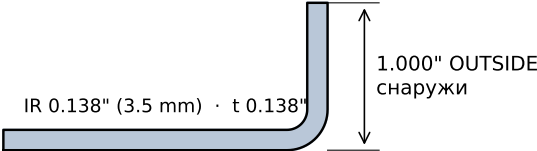
TOP VIEW (flanges up, toward viewer) / ВИД СВЕРХУ (борта вверх, к себе)



BEND 1 · long flange / длинный борт



BEND 2 · slant flange / косой борт



ISOMETRIC / ИЗОМЕТРИЯ

BEND TABLE / ТАБЛИЦА ГИБОВ

#	edge / кромка	angle	dir / направл.	IR	flange length	finished outside
1	long edge / длинная	90°	UP / вверх	0.138"	12.869"	2.116"
2	slant edge / косая	90°	UP / вверх	0.138"	4.148"	1.000"

Order: any (bends do not touch). Порядок гибов любой – они не пересекаются.
Both flanges on the SAME side. Оба борта в одну сторону.
FL-1 (mirror): same blank, bend both flanges to the OTHER side. FL-1: та же заготовка, гнуть в другую сторону.

- CHECK AFTER BENDING / ПРОВЕРИТЬ ПОСЛЕ ГИБКИ:
- long flange outside height 2.116" (±1/32")

• slant flange outside height 1.000" (±1/32")

• center hole Ø1.024" (26 mm) is 1.750" from the outside of the long flange

• ear holes Ø0.709" (18 mm) both, are 5.000" apart (c-c) – they bolt to FR-2

FR-1 · side plate / боковая пластина

A36 HR steel 3.5 mm (0.138")

inch / дюймы

Sienna WAV · 2026-10-02

FINISHED PART, BEND TABLE / ГОТОВАЯ ДЕТАЛЬ, ГИБЫ

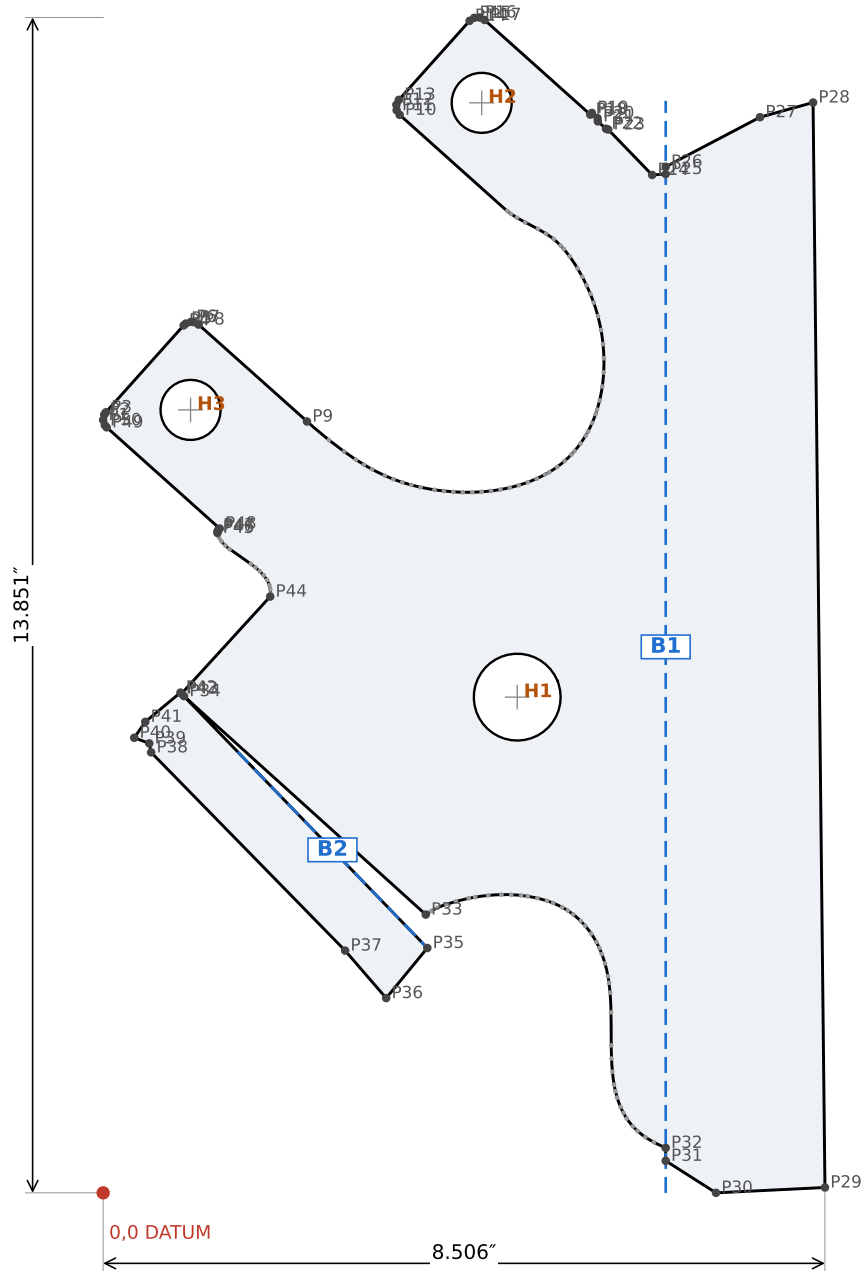
QTY 2 (FR-1 + FL-1 mirror)

cut ±0.010" · bend ±1°

SHEET 2 / 2

FR-1 — blank 1 of 1

Plate with ears — flat pattern (2 bends 90°) · blank 8.506″ × 13.851″ · HR steel sheet 0.138″ · 1 per side · FR + FL (mirror) = 2 per vehicle



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 0.138″ (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * bend 90° up on dashed lines

HOLES (centre X, Y from datum)

H1	Ø1.024″	X 4.880	Y 5.842
H2	Ø0.709″	X 4.461	Y 12.845
H3	Ø0.709″	X 1.030	Y 9.226

BENDS (bend centre line, from datum) IR = t = 0.138″

B1	(1) 90°	(6.630, -0.001) → (6.630, 12.869)
B2	(2) 90°	(1.584, 5.199) → (3.821, 2.885)

OUTLINE POINTS (X, Y from datum)

P1	0.000	9.108
P2	0.013	9.171
P3	0.032	9.198
P4	0.949	10.222
P5	0.974	10.244
P6	1.035	10.264
P7	1.068	10.261
P8	1.125	10.232
P9	2.402	9.090
P10	3.495	12.704
P11	3.459	12.758
P12	3.456	12.823
P13	3.485	12.880
P14	4.319	13.812
P15	4.373	13.847
P16	4.437	13.851
P17	4.495	13.822
P18	5.741	12.706
P19	5.760	12.727
P20	5.827	12.665
P21	5.830	12.627
P22	5.927	12.540
P23	5.952	12.531
P24	6.470	11.995
P25	6.629	12.007
P26	6.630	12.092
P27	7.741	12.676
P28	8.366	12.849
P29	8.506	0.063
P30	7.222	0.000
P31	6.630	0.377
P32	6.630	0.530
P33	3.802	3.279
P34	0.954	5.853
P35	3.821	2.885
P36	3.336	2.295
P37	2.852	2.855
P38	0.566	5.191
P39	0.544	5.297
P40	0.367	5.363
P41	0.497	5.551
P42	0.911	5.895
P43	0.924	5.883
P44	1.969	7.027
P45	1.347	7.777
P46	1.348	7.793
P47	1.354	7.808
P48	1.374	7.829
P49	0.041	9.021
P50	0.020	9.046

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.661	8.872
2.840	8.736
2.978	8.640

FR-1 — blank 1 of 1	HR steel sheet 0.138″	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate with ears — flat pattern (2 bends 90°)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010″ · bends ±1° · IR = t	SHEET 1 / 1

3.073	8.581
3.170	8.527
3.318	8.453
3.421	8.410
3.525	8.373
3.632	8.341
3.740	8.315
3.905	8.285
4.016	8.271
4.127	8.261
4.238	8.256
4.350	8.256
4.461	8.261
4.572	8.271
4.682	8.288
4.791	8.310
4.899	8.339
5.005	8.374
5.109	8.416
5.209	8.465
5.258	8.493
5.351	8.554
5.438	8.624
5.519	8.701
5.556	8.743
5.626	8.830
5.687	8.924
5.715	8.972
5.763	9.073
5.804	9.177
5.822	9.230

5.851	9.338
5.883	9.502
5.899	9.668
5.902	9.835
5.891	10.003
5.868	10.168
5.831	10.331
5.782	10.490
5.742	10.594
5.696	10.695
5.646	10.796
5.591	10.894
5.498	11.036
5.428	11.123
5.391	11.164
5.309	11.239
5.220	11.303
5.123	11.359
4.973	11.436
4.876	11.491
4.806	11.536
4.762	11.570

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

6.490	0.596
6.389	0.657
6.297	0.728
6.242	0.783
6.171	0.875
6.131	0.942
6.097	1.012
6.057	1.122

FR-1 — blank 1 of 1	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate with ears — flat pattern (2 bends 90°)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

6.037	1.197
6.022	1.273
6.001	1.427
5.991	1.583
5.987	1.738
5.985	2.127
5.974	2.360
5.962	2.475
5.950	2.552
5.915	2.704
5.877	2.814
5.847	2.886
5.812	2.955
5.752	3.055
5.681	3.148
5.628	3.204
5.540	3.281
5.476	3.326
5.409	3.366
5.304	3.415
5.231	3.442
5.118	3.473
5.004	3.495
4.926	3.505

4.810	3.513
4.694	3.515
4.538	3.505
4.384	3.485
4.308	3.471
4.157	3.434
4.045	3.400
3.936	3.358
3.867	3.323

FREE-FORM CURVE 3 (grey dots, follow DXF) — points X, Y

1.972	7.061
1.970	7.111
1.959	7.161
1.931	7.222
1.892	7.277
1.846	7.326
1.795	7.370
1.741	7.411
1.491	7.585
1.440	7.628
1.394	7.678
1.367	7.720
1.353	7.752

FR-1 — blank 1 of 1	HR steel sheet 0.138"	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate with ears — flat pattern (2 bends 90°)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1